DRUGS · ANTIFUNGALS AND ANTIVIRALS

Amphotericin B

Amphotericin B is a polyene that binds ergosterol and punches holes in fungal membranes. Liposomal amphotericin is the first-line drug for mucormycosis, the rapidly fatal angioinvasive fungal infection of the sinuses, orbit and brain seen in diabetic ketoacidosis, neutropenia, transplantation, high-dose steroids and, in recent years, after COVID-19. It is also the empirical choice for any acute invasive fungal sinusitis before the organism is identified.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

AMPHOTERICIN B

	SUMMARY
Class	Polyene antifungal
How it works	Binds ergosterol in the fungal cell membrane, forming pores that cause leakage and death; fungicidal
Spectrum	Mucorales (Rhizopus, Mucor, Lichtheimia), Aspergillus, Candida, Cryptococcus and most pathogenic fungi
ENT uses	Rhino-orbito-cerebral mucormycosis; acute invasive fungal sinusitis of unknown cause; severe invasive aspergillosis when azoles cannot be used
Dose	Liposomal amphotericin 5 to 10 mg/kg once daily intravenously for mucormycosis (3 mg/kg for other indications); conventional (deoxycholate) amphotericin is far more toxic and is avoided where possible
Pregnancy	Use if needed (liposomal preferred)

2 CHAPTER TWO Using it in ENT

Indications.

2.1 Mucormycosis

Suspect it in a diabetic with ketoacidosis or an immunocompromised patient with facial pain, swelling, numbness, nasal discharge or black eschar on the turbinate or palate, ophthalmoplegia or vision loss. Start **liposomal amphotericin at once**, before confirmation if the suspicion is high, and perform **urgent, repeated surgical debridement** to bleeding tissue, which may include orbital exenteration. Correct hyperglycaemia and acidosis, stop iron chelation with deferoxamine (a siderophore that feeds the fungus), and reduce immunosuppression. Step down to posaconazole or isavuconazole once stable. Mortality is high (often 30% to 50% or more).

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- **Nephrotoxicity** (less with liposomal forms): monitor creatinine daily; hydrate with saline.
- **Hypokalaemia and hypomagnesaemia** from tubular leak: monitor and replace.
- Infusion reactions: fever, rigors, hypotension; slowing the infusion and premedication help.
- Anaemia with long courses.

3.2 Interactions

Other nephrotoxins (aminoglycosides, ciclosporin, vancomycin), drugs affected by

hypokalaemia (digoxin, QT-prolonging drugs).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Risk factors and signs of **mucormycosis**.
- The triad of treatment: amphotericin, debridement, correcting the host.
- Renal and electrolyte monitoring.

4.2 Common pitfalls

COMMON PITFALLS

- Waiting for culture before starting amphotericin.
- Giving deferoxamine to a patient with suspected mucormycosis.

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